

p-series Test

p-Series Test for Infinite Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$

a) If $p > 1$ then the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges.

b) If $p \leq 1$ then the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges.

1) $\sum_{n=1}^{\infty} \frac{1}{n^3}$

a) $\int_1^{\infty} \frac{1}{x^3} dx =$ _____

b) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges.

c) Use the p -series Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges.

2) $\sum_{n=1}^{\infty} \frac{1}{n^{1/4}}$

a) $\int_1^{\infty} \frac{1}{x^{1/4}} dx =$ _____

b) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n^{1/4}}$ diverges.

c) Use the p -series Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n^{1/4}}$ diverges.

$$3) \sum_{n=1}^{\infty} \frac{1}{\sqrt[7]{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{1/7}}$$

Use the p -series Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n^{1/7}}$ diverges.

$$4) \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$$

Use the p -series Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$ converges.

5) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^{7/2}}$

a) Use the Integral Test to explain why the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^{7/2}}$ converges or diverges.

b) Use the p -series Test to explain why the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^{7/2}}$ converges or diverges

6) Does the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 4}$ converges or diverges?

$$\text{Note: } n^2 + 4 > n^2 \Rightarrow \frac{1}{n^2 + 4} < \frac{1}{n^2}.$$

$$\text{Hence, } \sum_{n=1}^{\infty} \frac{1}{n^2 + 4} < \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

a) What does the p -series test say about the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$?

b) What can you say about the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 4}$?

7) Does the series $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} + \frac{1}{n^3} \right)$ converges or diverges?

Note: $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} + \frac{1}{n^3} \right) = \sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{n^3} \right).$

a) Using the p -series Test, what can we say about the series $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} \right)$?

b) Using the p -series Test, what can we say about the series $\sum_{n=1}^{\infty} \left(\frac{1}{n^3} \right)$?

c) Explain why the series $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} + \frac{1}{n^3} \right)$ converges or diverges?

8) Does the series $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt[3]{5n^2}} \right)$ converges or diverges?

$$\text{Note: } \sum_{n=1}^{\infty} \left(\frac{1}{\sqrt[3]{5n^2}} \right) = \sum_{n=1}^{\infty} \left(\frac{1}{\sqrt[3]{5} \cdot \sqrt[3]{n^2}} \right) = \sum_{n=1}^{\infty} \left(\frac{1}{\sqrt[3]{5}} \cdot \frac{1}{n^{2/3}} \right) = \frac{1}{\sqrt[3]{5}} \cdot \sum_{n=1}^{\infty} \left(\frac{1}{n^{2/3}} \right)$$

a) Using the p -series Test, what can we say about the series $\sum_{n=1}^{\infty} \left(\frac{1}{n^{2/3}} \right)$?

b) Using the p -series Test, what can we say about the series $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt[3]{5n^2}} \right)$?

9) Does the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 4}$ converges or diverges?

$$\text{Note: } n^2 + 4 > n^2 \Rightarrow \frac{1}{n^2 + 4} < \frac{1}{n^2} \Rightarrow \frac{\sqrt{n}}{n^2 + 4} < \frac{\sqrt{n}}{n^2} = \frac{1}{n^{3/2}}$$

$$\text{Hence, } \sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 4} < \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}.$$

a) What does the p -series test say about the series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$?

b) What can you say about the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 4}$?

10) Does the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + 4}$ converges or diverges?

Note: $2\sqrt{n} \geq \sqrt{n} + 4$ for $n \geq 4$. Look at graphs of $f(x)$ and $g(x)$.

$$\text{Hence, } \frac{1}{2\sqrt{n}} \leq \frac{1}{\sqrt{n} + 4} \text{ for } n \geq 4 \quad \Rightarrow \quad \sum_{n=1}^{\infty} \frac{1}{2\sqrt{n}} \leq \sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + 4}$$

$$\text{Now } \sum_{n=1}^{\infty} \frac{1}{2\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{2} \cdot \frac{1}{\sqrt{n}} = \frac{1}{2} \cdot \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}.$$

a) What does the p -series test say about the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$?

b) What does the p -series test say about the series $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n}} = \frac{1}{2} \cdot \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$?

c) What can you say about the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + 4}$?



